Full readout design

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1 Global optimum for the full-output Fisher information

The model is the specified scalar eigenmode with unknown nonzero real coherent amplitude a, mean photon number $E = a^2$, and two calibrated scales. Put $x = s\lambda$, $y = t\lambda$, and

$$f(u) = ue^{-u}, \qquad z(u) = u(1-u)e^{-u}, \qquad v = (f(x), f(y), \sqrt{1 - f(x)^2 - f(y)^2}).$$

All three output modes are accessible with phase-calibrated homodyne means $\sqrt{2}av$ and covariance $I/2$. Since $v^T v = 1$, the Gaussian Fisher matrix has $\mathcal{J}_{a\lambda} = 4av^T v' = 0$, and

$$\mathcal{J}_{\lambda|a} = \frac{E}{\lambda^2} C_F(x, y), \qquad C_F = 4\left[z(x)^2 + z(y)^2 + \frac{(f(x)z(x) + f(y)z(y))^2}{1 - f(x)^2 - f(y)^2}\right].$$

The same value is the coherent-state QFI for this real displacement family: the squared overlap of nearby coherent states is $\exp[-E\|dv\|^2]$, giving $4E\|v'\|^2$. This concerns the specified family and available outputs, not all probes and measurements.

Theorem 1. *Over all $x, y > 0$, C_F has a unique global maximum at $x = y = r$, where $r > 3/2$ is the unique upper root of*

$$F(r) = r^2 - 3r + 1 + 2r^3 e^{-2r} = 0.$$

The maximum and maximizing scales are

$$r = 2.52128156583213785047\ldots, \qquad C_F^{\max} = 8r^3 e^{-2r} = 0.82793585316582672742\ldots.$$

Proof. Write $P(u) = f(u)^2$, $Z(u) = z(u)^2$ and $g(u) = Z(u)/(1 - 2P(u))$. The denominators are positive because $P(u) \le e^{-2} < 1/2$. Let $m = \sup_{u>0} g(u)$. The function is continuous, vanishes at 0, 1 and infinity in the relevant limits, and is positive elsewhere, so the supremum is attained at an interior point away from 1.

Cauchy–Schwarz gives

$$\frac{C_F(x, y)}{4} \le \frac{Z(x) + Z(y)}{1 - P(x) - P(y)}.$$

Since $Z(u) \le m(1 - 2P(u))$, adding the two inequalities gives $C_F(x, y) \le 8m$. Equality holds at $x = y = u$ whenever $g(u) = m$. Conversely, equality in the Cauchy step requires the vectors $(f(x), f(y))$ and $(z(x), z(y))$ to be proportional. Because f is strictly positive on this domain and $z(u)/f(u) = 1 - u$, equality requires $x = y$. It remains to prove that g has exactly one global maximizer.

Its derivative factors as

$$g'(u) = \frac{-2u(u-1)e^{-2u}F(u)}{(1 - 2u^2 e^{-2u})^2}, \qquad F'(u) = (2u - 3)(1 - 2u^2 e^{-2u}).$$

Thus F strictly decreases to $3/2$ and strictly increases thereafter. Since $F(0) = 1$, $F(1) = -1 + 2/e^2 < 0$, and $F(u) \to \infty$, it has exactly one root in $(0, 1)$ and exactly one in $(3/2, \infty)$. The derivative signs, together with the zero endpoint values of g, make these the unique maxima on the two branches.

To compare the branch maxima without trusting a numerical optimization, for $0 < u < 1$ use $u^2(1-u)^2 \le 1/16$ and $e^{-2u} < 1$. Since $e > 8/3$,

$$g(u) < \frac{1/16}{1 - 2/e^2} < \frac{2}{23}.$$

On the upper branch, evaluating at $u = 5/2$ and using $e < 11/4$ gives

$$g(5/2) = \frac{225/16}{e^5 - 25/2} > \frac{225/16}{(11/4)^5 - 25/2} = \frac{4800}{49417} > \frac{2}{23}.$$

The elementary bounds on e follow from its series: the sum through $1/4!$ is $65/24 > 8/3$, and the remaining tail is less than $1/100$, giving $e < 65/24 + 1/100 < 11/4$. Therefore the upper critical point is the unique global maximum of g, and equality in the earlier bound forces $x = y = r$. Finally, $F(r) = 0$ implies $(1-r)^2 = r(1-2r^2e^{-2r})$, so $g(r) = r^3e^{-2r}$. This proves the stated coefficient. □

Symbolic differentiation, 80-digit root calculations and a separate two-dimensional numerical optimization agree. The numerical search is corroboration, not the proof of globality. The reported settings are locally tuned in the dimensionless coordinates $s\lambda, t\lambda$; an unknown λ requires a nominal estimate or an adaptive design to implement them.

The earlier two-monitored-output contract still has its different global optimum $(0.4121145891, 2.7010117643)$ and coefficient 0.4780676701. Full-output access raises the best coefficient by about 73.2% within these models. It changes the physical readout contract and is not free information obtainable from discarded output.